

MAIN PROJECT

Bank-Customer-Churn-Analysis

1. Project Overview and Objective:

This project involves cleaning, transforming, and analyzing bank customer data using **Python and data analysis techniques** to identify patterns and factors related to customer churn. The dataset contains information such as **credit score, age, geography, balance, number of products, activity status, and customer exit status**

The main objective is to analyze customer behavior and identify the key factors associated with customers leaving the bank. The analysis will help generate **meaningful business insights and recommendations** that can support the bank in improving customer retention.

2. Data Sources:

Source Description and Timeline Hugging Face / Bank Customer Churn / <https://huggingface.co/datasets/praneeth232/bank-customer-churn>

Domain: Banking Churn.

3. Problem Statement:

- **Identify the key factors** that influence customers to leave the bank.
- **Analyze customer behavior** based on factors such as age, geography, balance, credit score, and product usage.
- **Compare exited and retained customers** to understand the patterns associated with customer churn.

- **Provide business insights and recommendations** that can help the bank improve customer retention and reduce churn.

4.Attribute (Column/Features):

<u>Attribute Name</u>	<u>Data Type</u>	<u>Description</u>
CustomerId	Integer	Unique identification number for each customer
Surname	String (Text)	Last name of the customer
CreditScore	Numeric (Float)	Credit score of the customer
Geography	Categorical	Country/region where the customer is located
Age	Numeric (Float)	Age of the customer in years
Tenure	Integer	Number of years the customer has been with the bank
Balance	Numeric (Float)	Bank account balance of the customer
NumOfProduct	Numeric (Float)	Number of banking products used by the customer
HasCrCard	Numeric / Binary	Indicates whether the customer has a credit card
IsActiveMember	Numeric / Binary	Indicates whether the customer is an active bank member
EstimatedSalary	Numeric (Float)	Estimated annual salary of the customer
Exited	Integer / Binary	Indicates whether the customer left the bank (1 = Yes, 0 = No)

5.Tools & Technologies:

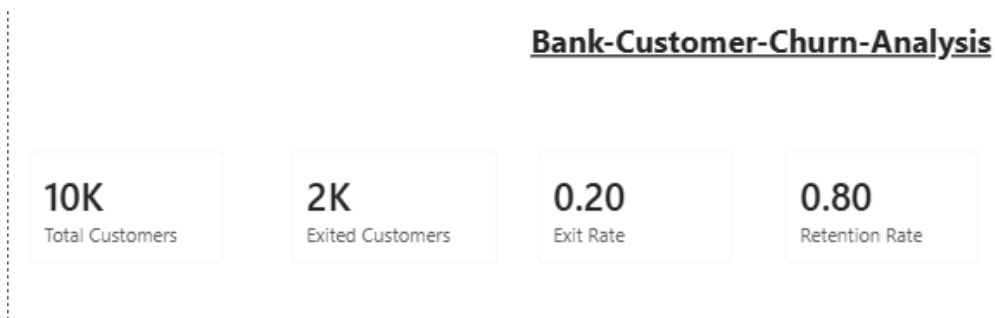
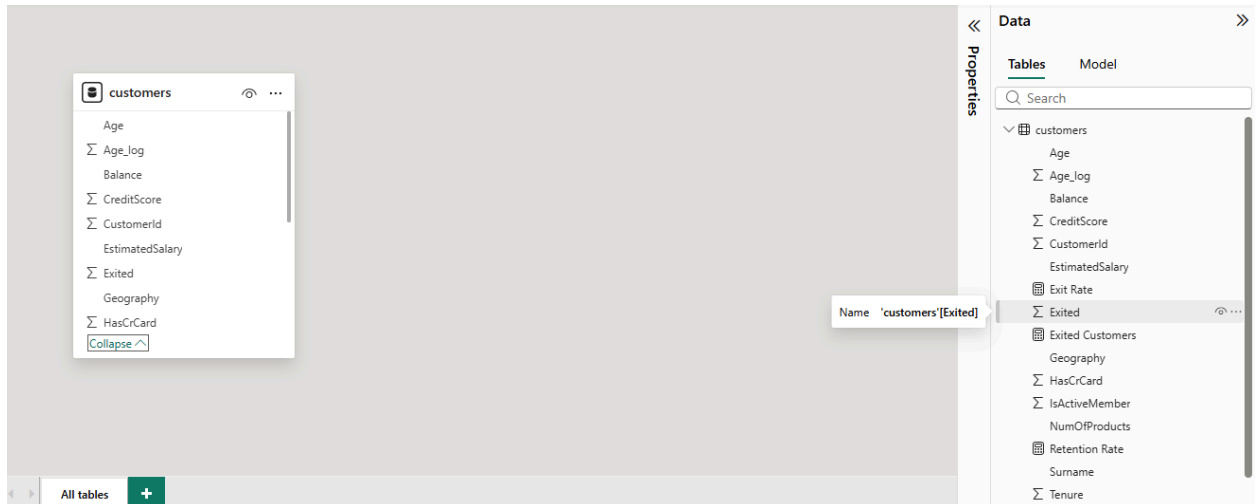
- **Python:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6.Data Pre-processing (Python/Power Query):

- **Data Cleaning & Transformation:** Checked the dataset for missing values and handled them appropriately. Removed duplicate records and corrected inconsistent or unnecessary data. Standardized the data formats and prepared the dataset for analysis.
- **Missing Value Handling:** Identified missing values in the dataset and handled them using suitable methods to ensure the data was complete and ready for analysis.
- **Data Validation:** Checked the columns for incorrect, unusual, or inconsistent values and validated the data types and ranges before performing analysis.
- **Outlier & Skewness Analysis:** Analyzed numerical columns for skewness and examined the distribution of the data. Appropriate transformations were applied where required to improve the quality of the dataset.
- **Data Transformation:** Created required calculated fields and transformed existing columns into suitable formats for analysis and visualization.
- **Filtering & Sorting:** Filtered and sorted the data to identify relevant customer segments and support further analysis.
- **Pivot Tables:** Created Pivot Tables to summarize customer information and identify patterns such as customer exit rates by geography, age group, active status, and number of products.
- **Power BI Preparation:** Prepared the cleaned and transformed dataset for Power BI dashboard development and business analysis.
- **Fact and Dimension Tables:** The dataset was kept as a single cleaned analytical table because the current project is focused on customer churn analysis. A separate Fact-Dimension structure was not necessary for this project.

7.Data Modelling and DAX (Power BI):

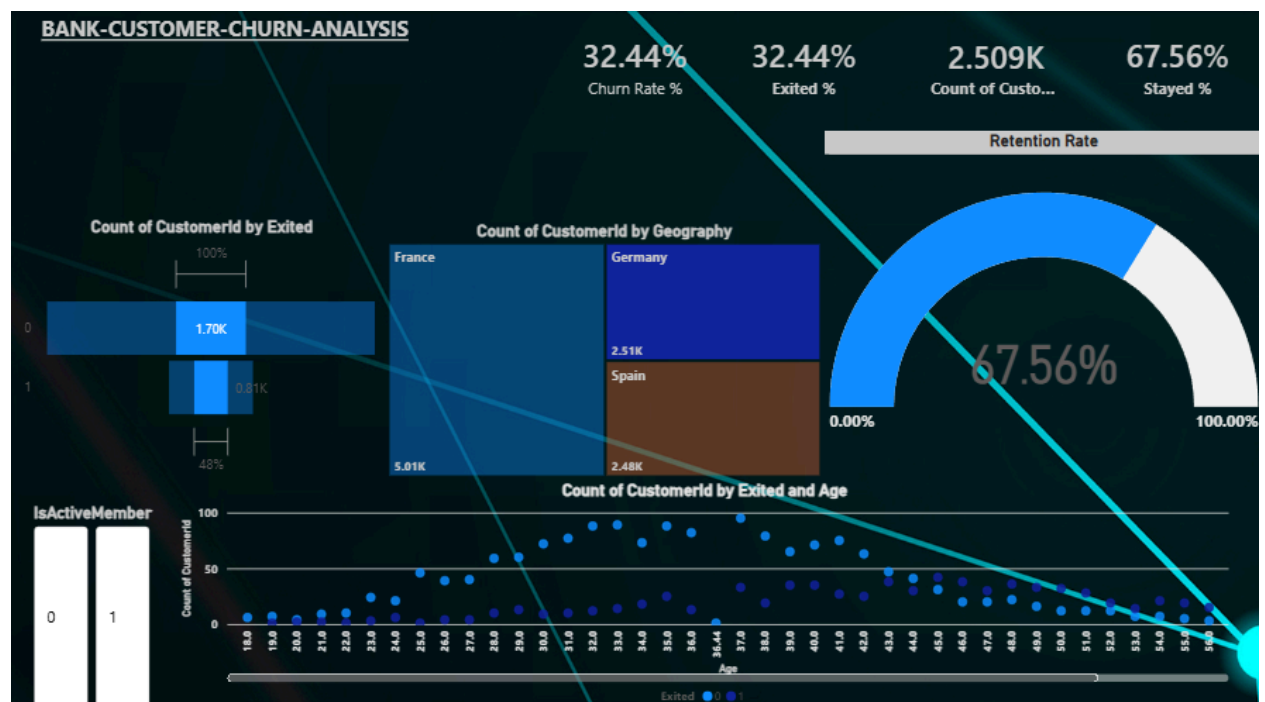
- **Structured the cleaned customer dataset into a single analytical table (**customers**) containing customer attributes such as Age, Geography, Balance, NumOfProducts, IsActiveMember, and Exited.**
- **Created DAX measures such as Exit Rate, Exited Customers, and Retention Rate to support customer churn analysis and Power BI dashboard visualizations.**



8. Analysis and Visualizations (Power BI):

Dashboard Features:

- **Customer Churn Analysis:** The dashboard analyzes customer churn by comparing **churned and retained customers**, helping identify the overall customer retention situation.
- **Geographical Analysis:** Customers are analyzed across **France, Germany, and Spain** to compare customer distribution and churn patterns between different countries.
- **Customer Behaviour Analysis:** The visualizations analyze factors such as **Tenure, Balance, Number of Products, Credit Card ownership, and Active Membership** to understand customer characteristics.
- **Interactive Visualization:** **Slicers for Geography and Tenure**, along with charts such as bar charts, pie/donut charts, line charts, and a matrix, allow users to interactively explore the data and identify important patterns.



9.Insights and Conclusions:

1. Descriptive Analysis — What happened?

- About **20.4% of customers exited**, while around **79.6% stayed**.
- **Germany has the highest exit rate (~32%)**, compared with France and Spain (~16–17%).
- Customers who exited are generally **older** and have a **higher average balance** than customers who stayed.

2. Diagnostic Analysis — Why did it happen?

- **Inactive customers are more likely to leave**: about **27% exited**, compared with **14% of active members**.
- Age is an important factor: customers aged **40–60 have a much higher exit rate**.
- Customers with **3 or more products show very high exit rates**, so this group needs attention.

3. Predictive Analysis — What may happen?

- Customers who are **older, inactive, and from Germany** are more likely to leave in the future.
- Customers with **high balances and multiple products** may also have a higher risk of exiting.

4. Prescriptive Analysis — What should we do?

- Focus retention offers on **inactive and older customers**, especially in **Germany**.
- Contact customers with **3+ products** and understand their problems before they leave.
- Encourage inactive customers to use banking services through **personalized offers or engagement campaigns**.

10.Future Enhancements:

****Target inactive customers:**** Provide personalized offers and regular communication to encourage inactive customers to use their accounts.

****Focus on older customers:**** Introduce simple banking services, dedicated support, and personalized assistance for older customers.

****Improve services in high-churn regions:**** Analyze why customers in Germany are leaving and improve products or services based on their needs.

Reduce customer dissatisfaction: Collect regular customer feedback and address common complaints quickly.

****Personalized offers:**** Provide suitable loans, savings plans, or other services based on each customer's profile and usage.

Monitor high-risk customers: Identify customers who are likely to leave and contact them before they churn.

****Improve customer engagement:**** Send useful notifications, rewards, and loyalty benefits to encourage customers to remain active.

10.Conclusion:

- The main factors related to customer exit are age, activity status, geography, and number of products. The bank should focus mainly on older and inactive customers, especially those in Germany, to improve customer retention.

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